

ECON 609: Particle Filter

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Lecture # 7

From Linear to Nonlinear Models

- ▶ While most dynamic models are inherently nonlinear, the nonlinearities are often small and decision rules are approximately linear.
- ▶ One can add certain features that generate more pronounced nonlinearities.
 - ▶ stochastic volatility
 - ▶ markov switching coefficients
 - ▶ asymmetric adjustment costs
 - ▶ occasionally binding constraints

From Linear to Nonlinear Models

- ▶ Linear dynamic model leads to

$$y_t = \Psi_0(\theta) + \Psi_1(\theta)s_t + u_t, \quad u_t \sim N(0, \Sigma_u)$$

$$s_t = \Phi_1(\theta)s_{t-1} + \Phi_\epsilon(\theta)\epsilon_t, \quad \epsilon_t \sim N(0, \Sigma_\epsilon)$$

- ▶ Nonlinear dynamic model leads to

$$y_t = \Psi(s_t; \theta) + u_t, \quad u_t \sim F_u(\cdot; \theta)$$

$$s_t = \Phi(s_{t-1}, \epsilon_t; \theta), \quad \epsilon_t \sim F_\epsilon(\cdot; \theta)$$

Some Comments

- ▶ Recall: Random Walk Metropolis-Hastings
- ▶ We can conduct Bayesian estimation using the posterior.
- ▶ We need to evaluate the likelihood given a parameter θ .
- ▶ For linear models: Kalman filter
- ▶ For nonlinear models: particle filter

Particle Filter

- ▶ There are many particle filters (PF)...
- ▶ We will focus on three types:
 - (1) Bootstrap PF
 - (2) Generic PF
 - (1) Conditionally-optimal PF

Filtering - General Idea

- ▶ State-space representation of nonlinear dynamic model
 - ▶ Measurement equation: $y_t = \Psi(s_t; \theta) + u_t, \quad u_t \sim F_u(\cdot; \theta)$
 - ▶ State transition equation: $s_t = \Phi(s_{t-1}, \epsilon_t; \theta), \quad \epsilon_t \sim F_\epsilon(\cdot; \theta)$
- ▶ Likelihood function: $p(Y_{1:T}|\theta) = \prod_{t=1}^T p(y_t|Y_{1:t-1}, \theta)$
- ▶ Iterations:
 - ▶ Initialization at time $t - 1$: $p(s_{t-1}|Y_{1:t-1}, \theta)$
 - ▶ Forecasting t given $t - 1$:
 - (1) Transition equation:
$$p(s_t|Y_{1:t-1}, \theta) = \int p(s_t|s_{t-1}, Y_{1:t-1}, \theta)p(s_{t-1}|Y_{1:t-1}, \theta)ds_{t-1}$$
 - (2) Measurement equation:
$$p(y_t|Y_{1:t-1}, \theta) = \int p(y_t|s_t, Y_{1:t-1}, \theta)p(s_t|Y_{1:t-1}, \theta)ds_t$$
 - ▶ Updating with Bayes theorem. Once y_t becomes available:

$$p(s_t|Y_{1:t}, \theta) = p(s_t|y_t, Y_{1:t-1}, \theta) = \frac{p(y_t|s_t, Y_{1:t-1}, \theta)p(s_t|Y_{1:t-1}, \theta)}{p(y_t|Y_{1:t-1}, \theta)}$$

Bootstrap Particle Filter

(1) Initialization: Draw the initial particles from the distribution

$s_0^j \sim_{i.i.d} p(s_0)$ and set $W_0^j = 1, j = 1, \dots, M$.

(2) Recursion. For $t = 1, \dots, T$:

- Forecasting s_t : Propagate the period $t - 1$ particles $\{s_{t-1}^j, W_{t-1}^j\}$ by iterating the state-transition equation forward:

$$\tilde{s}_t^j = \Phi(s_{t-1}^j, \epsilon_t^j; \theta), \quad \epsilon_t^j \sim F_\epsilon(\cdot; \theta)$$

An approximation of $\mathbb{E}[h(s_t) | Y_{1:t-1}, \theta]$ is given by

$$\hat{h}_{t,M} = \frac{1}{M} \sum_{j=1}^M h(\tilde{s}_t^j) W_{t-1}^j$$

Bootstrap Particle Filter

(1) Initialization.

(2) Recursion. For $t = 1, \dots, T$:

- ▶ Forecasting s_t .
- ▶ Forecasting y_t . Define incremental weights $\tilde{w}_t^j = p(y_t | \tilde{s}_t^j, \theta)$.

The predictive density $p(y_t | Y_{1:t-1}, \theta)$ can be approximated by

$$\hat{p}(y_t | Y_{1:t-1}, \theta) = \frac{1}{M} \sum_{j=1}^M \tilde{w}_t^j W_{t-1}^j$$

If the measurement errors $\sim N(0, \Sigma_u)$, then the incremental weights take the form

$$\tilde{w}_t^j = (2\pi)^{-n/2} |\Sigma_u|^{-1/2} \exp \left\{ -\frac{1}{2} (y_t - \Psi(\tilde{s}_t^j; \theta))' \Sigma_u^{-1} (y_t - \Psi(\tilde{s}_t^j; \theta)) \right\}$$

where n denotes the dimension of y_t .

Bootstrap Particle Filter

- (1) Initialization.
- (2) Recursion. For $t = 1, \dots, T$:
 - ▶ Forecasting s_t .
 - ▶ Forecasting y_t . Define incremental weights $\tilde{w}_t^j = p(y_t | \tilde{s}_t^j, \theta)$.
 - ▶ Updating. Define normalized weights

$$\tilde{W}_t^j = \frac{\tilde{w}_t^j W_{t-1}^j}{\frac{1}{M} \sum_{j=1}^M \tilde{w}_t^j W_{t-1}^j}$$

The larger the variance of the particle weights, the more inefficient the approximation. Define the effective sample size

$$\widehat{ESS}_n \equiv M / \left(\frac{1}{M} \sum_{j=1}^M (\tilde{W}_i^j)^2 \right)$$

Bootstrap Particle Filter

(1) Initialization.

(2) Recursion. For $t = 1, \dots, T$:

- ▶ Forecasting s_t .
- ▶ Forecasting y_t .
- ▶ Updating.
- ▶ Selection. If the effective sample size (ESS) falls below the half of the number of particles M , we do resampling.

Let $\{s_t^j\}_{j=1}^M$ denote M i.i.d. draws from a multinomial distribution characterized by support points and weights $\{\tilde{s}_t^j, \tilde{W}_t^j\}$ and set $W_t^j = 1$ for $j = 1, \dots, M$.
(Without resampling, (s_t^j, W_t^j) is set to be $(\tilde{s}_t^j, \tilde{W}_t^j)$.)

Likelihood Approximation

- ▶ The approximation of the log likelihood function is given by

$$\ln \hat{p}(Y_{1:T}|\theta) = \sum_{t=1}^T \ln \left(\frac{1}{M} \sum_{j=1}^M \tilde{w}_t^j W_{t-1}^j \right)$$

- ▶ Bootstrap filter needs non-degenerate $p(y_t|s_t, \theta)$ for incremental weights to be well-defined.

Generic Particle Filter

- (1) Initialization. Same as BSPF.
- (2) Recursion. For $t = 1, \dots, T$:
 - Forecasting s_t . Draw $\tilde{s}_t^j$ from density $g(\tilde{s}_t^j | s_{t-1}^j, \theta)$ and define

$$\omega_t^j = \frac{p(\tilde{s}_t^j | s_{t-1}^j, \theta)}{g(\tilde{s}_t^j | s_{t-1}^j, \theta)}$$

An approximation of $\mathbb{E}[h(s_t) | Y_{1:t-1}, \theta]$ is given by

$$\hat{h}_{t,M} = \frac{1}{M} \sum_{j=1}^M h(\tilde{s}_t^j) \omega_t^j w_{t-1}^j$$

- Forecasting y_t . Define incremental weights $\tilde{w}_t^j = p(y_t | \tilde{s}_t^j, \theta) \omega_t^j$. Then same as BSPF.
- Updating. Same as BSPF.
- Selection. Same as BSPF.

- (3) Likelihood approximation. Same as BSPF.

Adapting the Generic PF

- ▶ Conditionally-optimal importance distribution:

$$g(\tilde{s}_t | s_{t-1}^j) = p(\tilde{s}_t | y_t, s_{t-1}^j).$$

This is the posterior of s_t given s_{t-1}^j . Typically infeasible, but a good benchmark.

- ▶ Conditionally-linear models: do Kalman filter updating on s_t .

Example:

$$y_t = \Psi_0 + \Psi_1 s_t + u_t, \quad u_t \sim N(0, \Sigma_u),$$

$$s_t = \Phi_0 + \Phi_1 s_{t-1} + \Phi_\epsilon \epsilon_t, \quad \epsilon_t \sim N(0, \Sigma_\epsilon).$$

More on Conditionally-Linear Models

- ▶ With linear state-space representation,

$$s_t | (Y_{1:t}) \sim N(\bar{s}_{t|t}, P_{t|t}).$$

- ▶ Vector of means $\bar{s}_{t|t}$ and the covariance matrix $P_{t|t}$ are sufficient statistics for the conditional distribution of s_t .
- ▶ The Kalman filter forecasting step can be used to compute:

$$\tilde{s}_{t|t-1}^j = \Phi_0 + \Phi_1 s_{t-1}^j$$

$$P_{t|t-1}^j = \Phi_\epsilon P_{t-1|t-1}^j \Phi_\epsilon' + \Sigma_\epsilon$$

$$\tilde{y}_{t|t-1}^j = \Psi_0 + \Psi_1 \tilde{s}_{t|t-1}^j$$

$$F_{t|t-1}^j = \Psi_1 P_{t|t-1}^j \Psi_1' + \Sigma_u$$

More on Conditionally-Linear Models

- ▶ We can use the Kalman filter to update the information about s_t in view of the current observation y_t :

$$\begin{aligned}\tilde{s}_{t|t}^j &= \tilde{s}_{t|t-1}^j + P_{t|t-1}^j \Psi_1'(F_{t|t-1}^j)^{-1} (y_t - \bar{y}_{t|t-1}^j) \\ \tilde{P}_{t|t}^j &= P_{t|t-1}^j - P_{t|t-1}^j \Psi_1'(F_{t|t-1}^j)^{-1} \Psi_1 P_{t|t-1}^j.\end{aligned}$$

- ▶ The likelihood approximation is based on the incremental weights with

$$\omega_t^j = \frac{p(\tilde{s}_t | s_{t-1}^j)}{p(\tilde{s}_t | y_t, s_{t-1}^j)}$$

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